

NUMM Project Workflow

National Unified Material Master ? End-to-End System Workflow

1. Project Objective

NUMM is a national material harmonization and governance platform that brings together CPSE material records, AI-powered matching, reviewer-controlled approvals, inventory demand logic, SAP integration, and auditability into one operational workflow.

2. System Architecture

The platform is composed of three layers: React frontend, FastAPI backend, and AI matching service. The backend stores data in PostgreSQL and coordinates import, validation, approvals, inventory, requests, and integration flows. The AI service handles normalization, candidate retrieval, and scoring.

3. Core Workflow

The full workflow is: login -> dashboard -> import files -> validation and cleaning -> AI matching -> reviewer approval -> national material registry -> inventory and demand management -> request allocation -> SAP sync -> audit and reporting.

4. Data Intake and Import

Material and inventory files are uploaded from the frontend. The backend validates them, inspects records, previews the impact, and confirms successful imports. Import batches are tracked with idempotency and queue processing for reliability.

5. Data Cleaning and Validation

Descriptions are normalized and unit conflicts are corrected. Technical attributes are extracted. Missing data, incompatible categories, or risky specifications are flagged before matching is accepted for downstream decision-making.

6. AI Matching

The backend sends material records to the AI service. AI performs normalization, structured attribute extraction, candidate retrieval, semantic and fuzzy scoring, and returns recommended labels such as EXACT, NEAR_DUPLICATE, FUNCTIONAL_EQUIVALENT, or NO_MATCH.

7. Human-in-the-Loop Approval

AI is advisory only. Human reviewers confirm final mapping decisions. Functional equivalent mappings cannot be auto-approved. Safety-critical attributes such as grade, pressure, voltage, dimensions, and material conflict conditions require manual review.

8. National Material Registry

Approved mappings are stored as canonical national material records. This creates one national identity across multiple CPSE-specific legacy records. Mapping history and rollback decisions are preserved for traceability.

9. Inventory and Demand

The platform tracks cross-CPSE stock, material availability, demand, and operational balances. The system validates ownership, unit compatibility, and quantity semantics before moving to allocation.

10. Material Requests and Allocation

Users create material requests. The system previews allocations based on current stock and demand, validates request details, and then approves and fulfills eligible requests in a controlled, auditable flow.

11. SAP / ERP Sync

The backend supports SAP syncing through a connector boundary. The mock connector supports demo flows, while OData-based integration is used for configured SAP environments with health checks, delta handling, and reconciliation features.

12. Audit, Governance, and Reporting

Every major step is recorded in the audit trail. Governance controls protect against unsafe auto-approvals, mandatory review of critical conflicts, and inconsistent inventory behavior. Reporting and exports support operational oversight and data-quality review.

Primary Workflow Summary

Login -> Dashboard -> Import -> Validation -> AI Matching -> Reviewer Approval -> National Registry -> Inventory & Demand -> Requests -> SAP Sync -> Audit & Reporting

Key Safety Rules

- 1 AI recommends; human reviewer decides.
- 2 FUNCTIONAL_EQUIVALENT must never be auto-mapped without review.
- 3 Critical conflicts (grade, pressure, voltage, dimensions) block approval until reviewed.
- 4 Missing technical evidence prevents automatic approval.
- 5 Incompatible units and invalid demand records are rejected.
- 6 Maker-checker rules prevent self-approval and self-rejection.
- 7 Every approval and operational change is stored in the audit trail.

Implementation Scope

The implemented platform covers material import, validation, AI-assisted harmonization, approvals, national registry, inventory and demand control, request handling, procurement analytics, and ERP sync. Future work still includes deeper domain-specific model calibration, persistent vector-index scaleup, and broader production-grade ERP lifecycle management.

Appendix A ? Route and Feature Map

Frontend routes include Dashboard, Materials, Import, AI Recommendations, Comparison, National Materials, Approvals, Inventory, Procurement History, Requests, CPSE Management, Data Quality, Integrations, and Audit Trail.

Backend services include authentication, CPSE management, materials, imports, matching, approvals, inventory, requests, procurement, demand, integrations, audits, exports, and data quality enforcement.

AI service routes include batch matching, direct pair matching, and candidate retrieval for recommendation generation.